CLEAR-IR: Clarity-Enhanced Active Reconstruction of Infrared Imagery for Low-light Enhancement

Nathan Shankar[1,2], Marton Gonczy[2], Hujun Yin[2], and Pawel Ladosz[1,2,*]

[1] RAICo1 Research Group

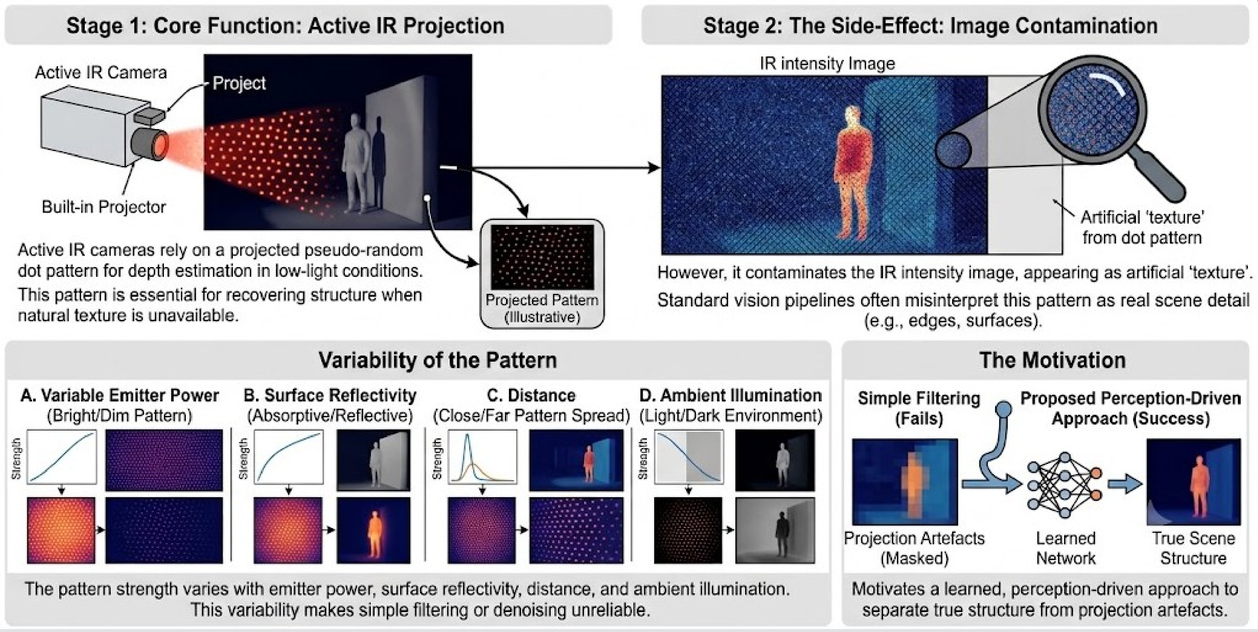
[2] The University of Manchester

1 Motivation

The Problem: Robots operating in completely dark or subterranean environments such as mine shafts, disaster zones, or nuclear facilities cannot rely on passive RGB vision, which fails entirely when no visible light reaches the sensor.

- **Passive RGB Limitations**: Conventional cameras fail under low-light conditions as dark current and readout noise dominate, while increasing exposure times introduces terminal motion blur in dynamic robotic tasks.
- **Active Visible Illumination Failures**: Onboard floodlighting or flash modules cause specular overexposure, harsh casting shadows, and suffer from heavy backscatter off airborne dust or particles and also results in extra setup complexity.
- **The Emitter Conflict**: Active IR sensing provides reliable structural signals regardless of ambient light, but its structured pseudo-random dot patterns heavily corrupt the raw stream. Standard downstream vision pipelines misinterpret these dots as real features, breaking tracking, segmentation, and localization loops.

2 The Emitter Pattern Problem



3 CLEAR-IR Architecture

Input: Emitter-corrupted active IR image.

Context Stream: U-Net encoder-decoder captures global scene structure.

Detail Stream: Full-resolution convolutions preserve fine texture and edges.

Fusion: Streams are fused additively and refined to produce the reconstructed RGB-styled image.

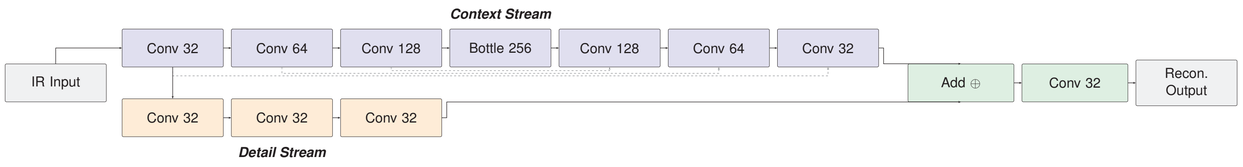
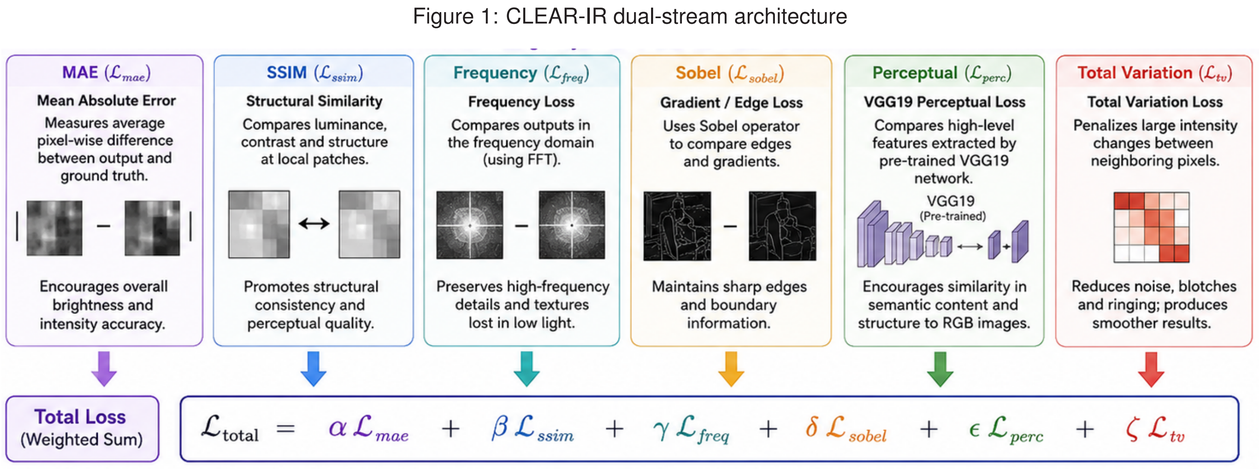


Figure 1: CLEAR-IR dual-stream architecture



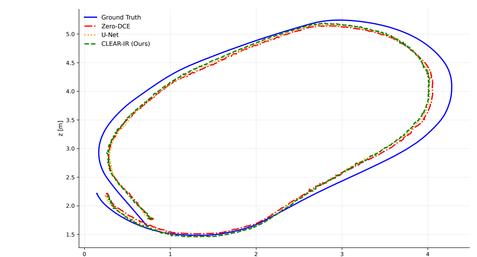
4 Visual SLAM Performance

Setup: Evaluated on a modified ORB-SLAM3 pipeline with SuperPoint tracking to accommodate CNN-smoothed gradients across three distinctive motion trajectories: Forward-Backward (FB), Square 1-lap (SQ1), and loop-closure Square 2-laps (SQ2).

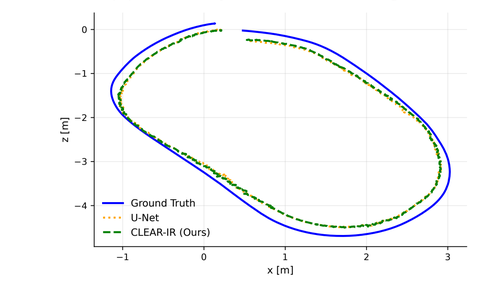
	Low Light RMSE (m)			Extreme Low Light RMSE (m)		
Method	FB	SQ1	SQ2	FB	SQ1	SQ2
Retinex [5]	**0.064**	0.163	0.168	DNI	DNI	DNI
CIDNet [2]	0.069	0.560	0.178	DNI	DNI	DNI
Raw IR	DNI	DNI	DNI	DNI	DNI	DNI
U-Net [4]	0.072	0.161	0.164	0.097	0.167	0.168
CLEAR-IR	0.070	0.164	**0.137**	**0.092**	**0.159**	**0.159**

Table 1: SLAM RMSE metrics (m). *DNI: Did Not Initialise.*

- **Low Light Analysis**: CLEAR-IR functions on par with SOTA RGB enhancement models, minimising cumulative multi lap translation drift down.
- **Extreme Low Light Evaluation**: All comparative RGB pipelines experience terminal tracking initialization dropouts (DNI). CLEAR-IR consistently minimises trajectory localisation errors over the structural baseline.



(a) SQ1 Trajectories – Low Light



(b) SQ1 Trajectories – Extreme Low Light

Figure 2: vSLAM Trajectory tracking

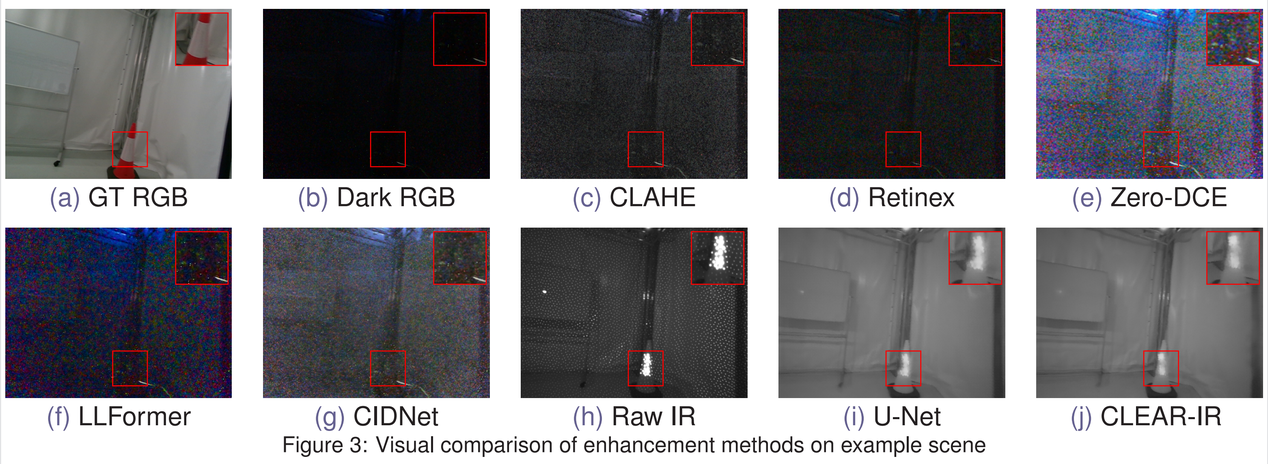
5 Object and ArUco Marker Detection

Raw IR is essentially unusable and YOLOv26 [1] fails to segment objects due to emitter dot corruption, and ArUco detection fails completely.

CLEAR-IR recovers **11**/**11** objects and **6**/**6** ArUco markers, matching or exceeding RGB-based SOTA methods that operate on a passively captured dark stream.

	Object Detection			ArUco	
Method	Detected	Missed	Mislabelled	Correct	Miss
RGB (passive dark stream)					
CIDNet [2]	1	9	0	**6**	**0**
IR (active stream)					
Raw IR	0	9	1	0	6
DeepMAO (ablation) [3]	2	8	2	0	6
U-Net (ablation) [4]	7	3	0	6	0
CLEAR-IR	**11**	**0**	2	**6**	**0**

Table 2: Detection performance; full comparison spans nine baselines in the paper.



(a) GT RGB (b) Dark RGB (c) CLAHE (d) Retinex (e) Zero-DCE

(f) LLFormer (g) CIDNet (h) Raw IR (i) U-Net (j) CLEAR-IR

Figure 3: Visual comparison of enhancement methods on example scene

6 Conclusions

Core Contribution:

- **CLEAR-IR** uses a dual-stream architecture to reconstruct clean, RGB-styled images from IR streams with active emitter pattern.
- Enables standard, RGB-trained perception pipelines to operate seamlessly in **complete darkness**.

Key Evaluation Results:

- **Downstream Tasks:** Matches leading RGB methods across object detection, ArUco tracking, and visual SLAM.
- **Zero Light:** The only approach alongside its U-Net ablation sustain reliable perception when RGB sensors completely collapse in extreme low light.

Future Work:

- Model quantization for embedded deployment.
- Validation in scattering environments such as dust and smoke).

References

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